🌐 Lab 02: Multiple VPC Networks

Build a Secure Google Cloud Network | GSP211

Overview

Created multiple custom VPC networks with firewall rules and VM instances, then tested connectivity to understand how VPC isolation works. Created a VM with multiple network interfaces to bridge across VPC networks.

Network Architecture

Network	Mode	Created By
default	Auto	GCP (pre-existing)
mynetwork	Auto	Pre-created for lab
managementnet	Custom	Console
privatenet	Custom	CLI

Objectives

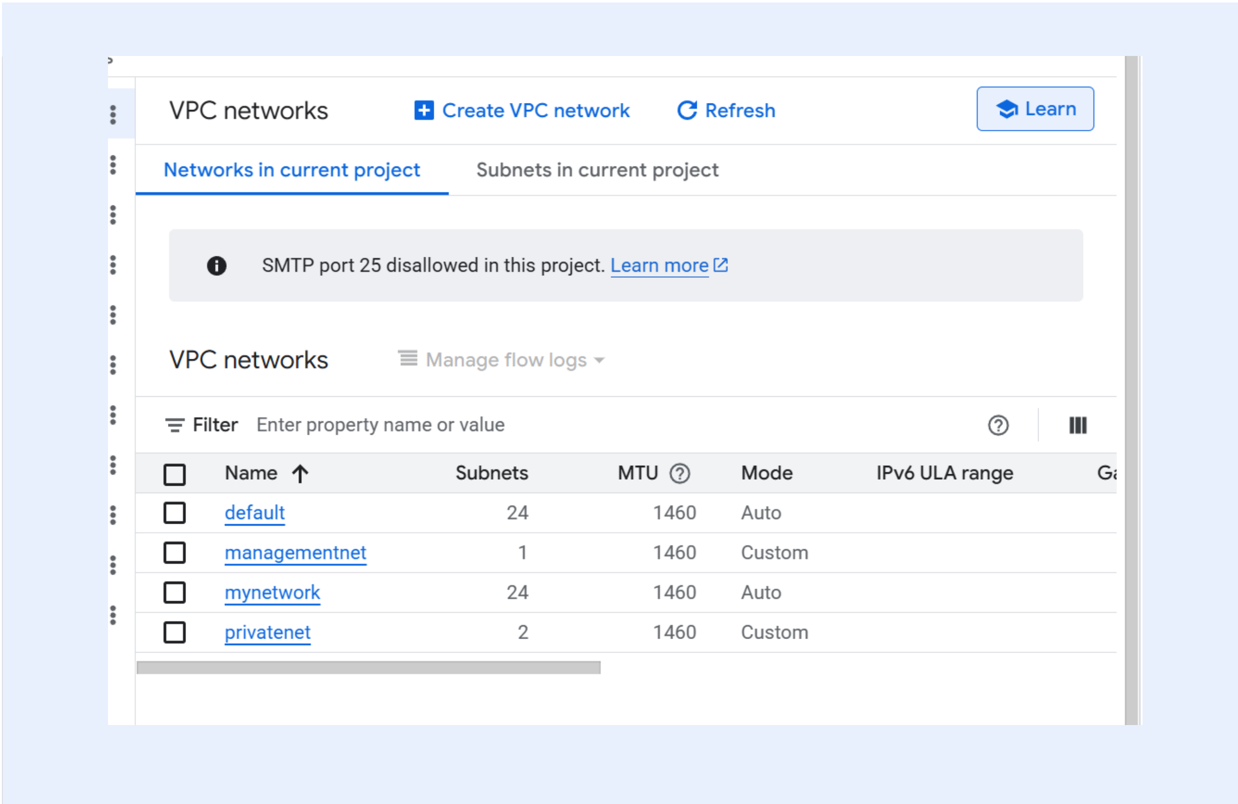
- Create custom mode VPC networks with firewall rules
- Create VM instances in different VPC networks
- Test connectivity across VPC networks using ping
- Create a VM with multiple network interfaces

Task 1 - Create Custom VPC Networks

Created managementnet via console and privatenet via CLI.

```
gcloud compute networks create privatenet --subnet-mode=custom
gcloud compute networks subnets create privatesubnet-1 --network=privatenet --
region=[REGION] --range=172.16.0.0/24
gcloud compute networks subnets create privatesubnet-2 --network=privatenet --
region=[REGION] --range=172.20.0.0/20
gcloud compute networks list
gcloud compute networks subnets list --sort-by=NETWORK
```

```
gcloud compute firewall-rules create privatenet-allow-icmp-ssh-rdp --
direction=INGRESS --priority=1000 --network=privatenet --action=ALLOW --
rules=icmp,tcp:22,tcp:3389 --source-ranges=0.0.0.0/0
```

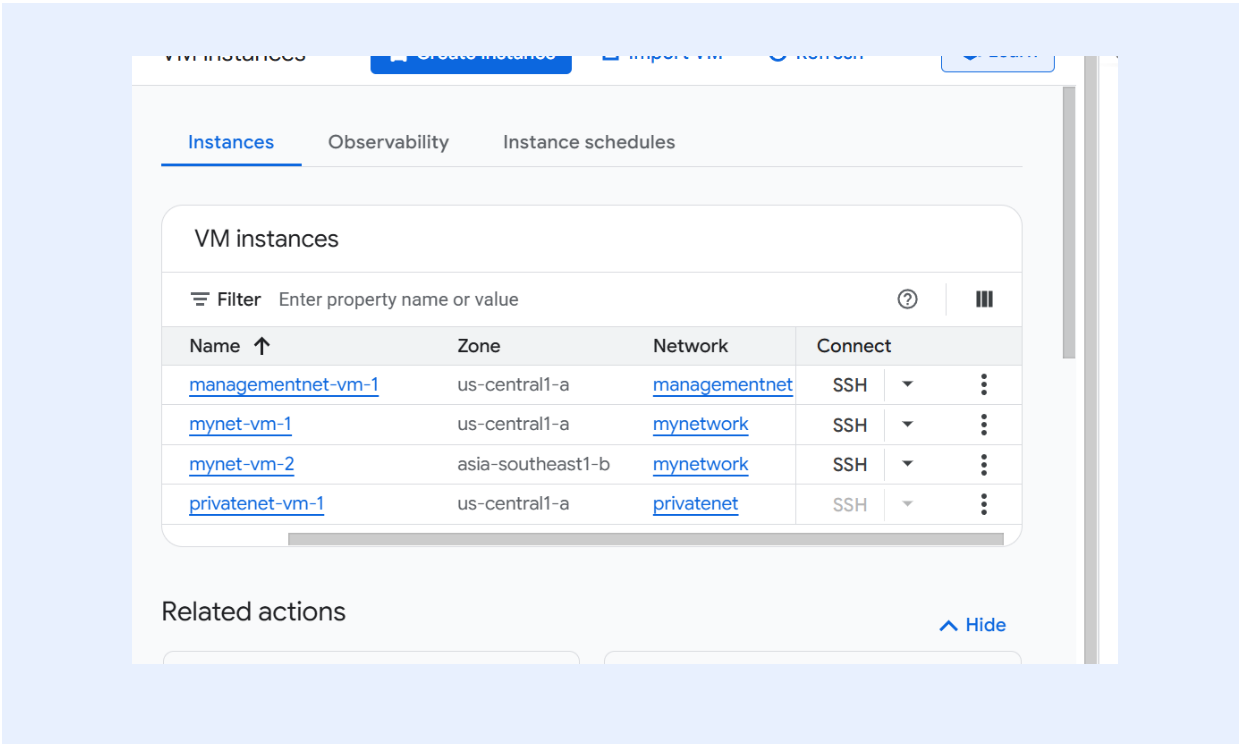

VPC networks
Create VPC network
Refresh
Learn
Networks in current project
Subnets in current project
SMTP port 25 disallowed in this project. Learn more
VPC networks
Manage flow logs
Filter
Enter property name or value
Name
Subnets
MTU
Mode
IPv6 ULA range
default 24 1460 Auto
managementnet 1 1460 Custom
mynetwork 24 1460 Auto
privatenet 2 1460 Custom

```
NAME: mynet-vm-2
ZONE: asia-southeast1-b
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.148.0.2
EXTERNAL_IP: 34.21.242.179
STATUS: RUNNING

NAME: managementnet-vm-1
ZONE: us-central1-a
MACHINE_TYPE: e2-micro
PREEMPTIBLE:
INTERNAL_IP: 10.130.0.2
EXTERNAL_IP: 34.173.30.35
STATUS: RUNNING

NAME: mynet-vm-1
ZONE: us-central1-a
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.128.0.2
EXTERNAL_IP: 34.16.126.188
STATUS: RUNNING

NAME: privatenet-vm-1
ZONE: us-central1-a
MACHINE_TYPE: e2-micro
PREEMPTIBLE:
INTERNAL_IP: 172.16.0.2
EXTERNAL_IP: 34.55.106.222
STATUS: RUNNING
```

Task 2 - Create VM Instances

Created managementnet-vm-1 via console and privatenet-vm-1 via CLI.

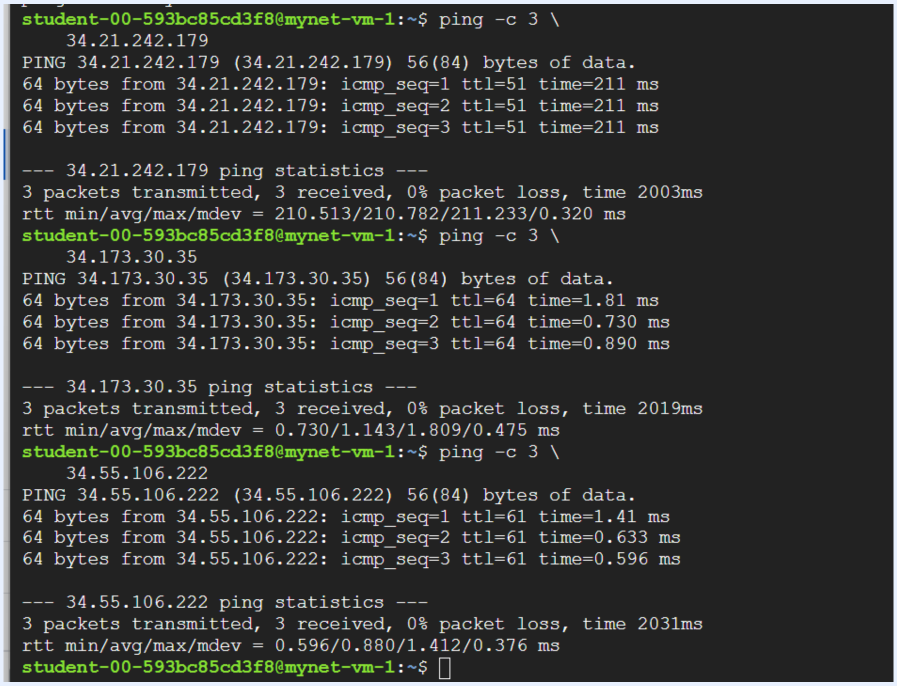
```
gcloud compute instances create privatenet-vm-1 --zone=[ZONE] --machine-type=e2-micro --subnet=privatesubnet-1
gcloud compute instances list --sort-by=ZONE
```



Task 3 - Test Connectivity Between VMs

Tested ping between VMs using external and internal IPs to understand VPC isolation.

- External IP pings: ALL worked regardless of VPC network
- Internal IP ping to mynet-vm-2: WORKED (same VPC network)
- Internal IP ping to managementnet-vm-1: FAILED (different VPC)
- Internal IP ping to privatenet-vm-1: FAILED (different VPC)



```
student-00-593bc85cd3f8@mynet-vm-1:~$ ping -c 3 \
    34.21.242.179
PING 34.21.242.179 (34.21.242.179) 56(84) bytes of data.
64 bytes from 34.21.242.179: icmp_seq=1 ttl=51 time=211 ms
64 bytes from 34.21.242.179: icmp_seq=2 ttl=51 time=211 ms
64 bytes from 34.21.242.179: icmp_seq=3 ttl=51 time=211 ms

--- 34.21.242.179 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 210.513/210.782/211.233/0.320 ms
student-00-593bc85cd3f8@mynet-vm-1:~$ ping -c 3 \
    34.173.30.35
PING 34.173.30.35 (34.173.30.35) 56(84) bytes of data.
64 bytes from 34.173.30.35: icmp_seq=1 ttl=64 time=1.81 ms
64 bytes from 34.173.30.35: icmp_seq=2 ttl=64 time=0.730 ms
64 bytes from 34.173.30.35: icmp_seq=3 ttl=64 time=0.890 ms

--- 34.173.30.35 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2019ms
rtt min/avg/max/mdev = 0.730/1.143/1.809/0.475 ms
student-00-593bc85cd3f8@mynet-vm-1:~$ ping -c 3 \
    34.55.106.222
PING 34.55.106.222 (34.55.106.222) 56(84) bytes of data.
64 bytes from 34.55.106.222: icmp_seq=1 ttl=61 time=1.41 ms
64 bytes from 34.55.106.222: icmp_seq=2 ttl=61 time=0.633 ms
64 bytes from 34.55.106.222: icmp_seq=3 ttl=61 time=0.596 ms

--- 34.55.106.222 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2031ms
rtt min/avg/max/mdev = 0.596/0.880/1.412/0.376 ms
student-00-593bc85cd3f8@mynet-vm-1:~$
```

Task 4 - Create VM with Multiple Network Interfaces

Created vm-appliance with 3 NICs connected to privatenet, managementnet, and mynetwork simultaneously.

- nic0: privatesubnet-1 (172.16.0.0/24)
- nic1: managementsubnet-1 (10.130.0.0/20)
- nic2: mynetwork (10.128.0.0/20)

```
sudo ifconfig
ip route
```

SSH-in-browser UPLOAD FILE DOWN

```
permitted by applicable law.
Creating directory '/home/student-00-593bc85cd3f8'.
student-00-593bc85cd3f8@vm-appliance:~$ sudo ifconfig
ens4: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1460
        inet 172.16.0.4  netmask 255.255.255.255  broadcast 0.0.0.0
        inet6 fe80::4001:acff:fe10:4  prefixlen 64  scopeid 0x20<link>
        ether 42:01:ac:10:00:04  txqueuelen 1000  (Ethernet)
        RX packets 7129  bytes 141033069 (134.4 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 3409  bytes 1471226 (1.4 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

ens5: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1460
        inet 10.130.0.3  netmask 255.255.255.255  broadcast 0.0.0.0
        inet6 fe80::4001:aff:fe82:3  prefixlen 64  scopeid 0x20<link>
        ether 42:01:0a:82:00:03  txqueuelen 1000  (Ethernet)
        RX packets 29  bytes 3362 (3.2 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 62  bytes 4903 (4.7 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

ens6: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1460
        inet 10.128.0.4  netmask 255.255.255.255  broadcast 0.0.0.0
        inet6 fe80::4001:aff:fe80:4  prefixlen 64  scopeid 0x20<link>
        ether 42:01:0a:80:00:04  txqueuelen 1000  (Ethernet)
        RX packets 2096  bytes 1323313 (1.2 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 226  bytes 25374 (24.7 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 60  bytes 53041 (51.7 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 60  bytes 53041 (51.7 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

student-00-593bc85cd3f8@vm-appliance:~$
```

```
64 bytes from 172.16.0.2: icmp_seq=1 ttl=64 time=1.58 ms
64 bytes from 172.16.0.2: icmp_seq=2 ttl=64 time=0.446 ms
64 bytes from 172.16.0.2: icmp_seq=3 ttl=64 time=0.294 ms

--- 172.16.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2031ms
rtt min/avg/max/mdev = 0.294/0.774/1.583/0.575 ms
student-00-593bc85cd3f8@vm-appliance:~$ ping -c 3 10.130.0.2
PING 10.130.0.2 (10.130.0.2) 56(84) bytes of data.
64 bytes from 10.130.0.2: icmp_seq=1 ttl=64 time=2.01 ms
64 bytes from 10.130.0.2: icmp_seq=2 ttl=64 time=0.380 ms
64 bytes from 10.130.0.2: icmp_seq=3 ttl=64 time=0.439 ms

--- 10.130.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2014ms
rtt min/avg/max/mdev = 0.380/0.943/2.010/0.754 ms
student-00-593bc85cd3f8@vm-appliance:~$ ping -c 3 10.128.0.2
PING 10.128.0.2 (10.128.0.2) 56(84) bytes of data.
64 bytes from 10.128.0.2: icmp_seq=1 ttl=64 time=1.77 ms
64 bytes from 10.128.0.2: icmp_seq=2 ttl=64 time=0.444 ms
64 bytes from 10.128.0.2: icmp_seq=3 ttl=64 time=0.325 ms

--- 10.128.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2030ms
rtt min/avg/max/mdev = 0.325/0.845/1.766/0.653 ms
student-00-593bc85cd3f8@vm-appliance:~$ ping -c 3 10.148.0.2
PING 10.148.0.2 (10.148.0.2) 56(84) bytes of data.

--- 10.148.0.2 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2056ms

student-00-593bc85cd3f8@vm-appliance:~$ ip route
default via 172.16.0.1 dev ens4 proto dhcp src 172.16.0.4 metric 100
10.128.0.0/20 via 10.128.0.1 dev ens6 proto dhcp src 10.128.0.4 metric 100
10.128.0.1 dev ens6 proto dhcp scope link src 10.128.0.4 metric 100
10.130.0.0/20 via 10.130.0.1 dev ens5 proto dhcp src 10.130.0.3 metric 100
10.130.0.1 dev ens5 proto dhcp scope link src 10.130.0.3 metric 100
169.254.169.254 via 172.16.0.1 dev ens4 proto dhcp src 172.16.0.4 metric 100
172.16.0.0/24 via 172.16.0.1 dev ens4 proto dhcp src 172.16.0.4 metric 100
172.16.0.1 dev ens4 proto dhcp scope link src 172.16.0.4 metric 100
student-00-593bc85cd3f8@vm-appliance:~$
```

What I Learned

- VMs in the same VPC network can communicate via internal IP regardless of zone or region
- VMs in different VPC networks cannot communicate via internal IP without VPC peering or VPN
- External IP communication works across VPC networks via the internet
- Auto mode networks create subnets in every region automatically
- Custom mode networks give full control over subnet creation
- A VM can have up to 8 NICs connecting it to multiple VPC networks
- The default route is associated with eth0 so traffic to unknown subnets leaves via eth0

Resources

GCP VPC Networking Docs: https://cloud.google.com/vpc/docs
Lab ID: GSP211